Online Supplement

Title: Claim-specific sample-size planning for binary hit-rate verification of detection capability

Companion: Main manuscript (clinical planning in the main text; mapping rules, lattice margins, audit ledger, and operating-characteristic tables here).

Main manuscript figures/tables: Fig. 1 (A–D), Fig. 2 (A–C); Tables 1–3.

Scope. Binary hit-rate claims only (each replicate scored detected or not). Blank-SD LoB/LoD, probit/logit concentration estimation, and EP26 quantitative lot tests are out of scope.

Numbering. Main-text tables are Tables 1–3. Supplement tables keep the labels cited in the main text (3e, 3f, 7b, 8, 9b, 9c) and are independent of that sequence. Supplement Table 7b is not main-text Table 3.

Rebuild. `python3 scripts/build_online_supplement.py`

S1. Claim-mapping rules (audit)

Four phrase rules were applied to 16 publicly accessible rows (Supplementary Table 7b).

1. A stated k-of-n or 19-of-20 pass rule → qualitative pass criteria.
2. A two-lot, two-material, or two-target contrast at the LoD, or a multi-lot detect grid → two-arm comparison. “Equivalent” without a numeric margin was treated as discrimination, not TOST.
3. A single-method 95% detection claim, LoD confirmation sentence, or low-level series without a pass rule or two-arm contrast → benchmark confirmation.
4. Hospital and published x/n tables without a two-arm sentence → benchmark confirmation.

Defaults when rates were unwritten: p0 = 0.95 for benchmark confirmation; (pA, pB) = (0.90, 0.80) for two-arm records (the Introduction planning pair, not a measured hit rate in those files). Replicate counts were not imputed. None of the 16 source texts stated an 80% power target.

S2. Lattice operating characteristics

The two-arm lattice is 1,225 equally spaced pairs with pA > pB on the integer 50–99% grid, scored with exact two-sided Fisher tests at n = 20 per arm (= 0.05).

- Cells: 1225
- Fraction with exact Fisher   > 0.80: 0.611
- Cells in which asymptotic pooled-z power exceeded exact Fisher by more than 0.05: 778 (this is the 778-cell count cited in the main text).

Fig. 1C displays n* on a 20–435 scale; that range is a plot scale (95%/90%   435) and does not cap 1-percentage-point cells.

Supplementary Table 3f. Exact Fisher underpowering and asymptotic n* by Δp band

delta_pp_band	n_cells	frac_beta_gt_0.8	median_power_n20	median_n_star_asymptotic	min_n_star	max_n_star
1–5	235	1.0	0.0252	2887.2	210.9	39239.3
6–10	210	1.0	0.0444	424.8	87.3	1085.0
11–20	345	0.8609	0.1236	116.0	37.2	319.2
>20	435	0.0138	0.488	31.6	11.2	83.9

Supplementary Table 3e. Asymptotic versus exact Fisher power at n = 20 (first 25 cells)

p_a	p_b	delta_pp	power_exact_fisher_n20	power_asymptotic_n20	asymp_minus_exact	n_star_asymptotic	beta_gt_0.8
0.51	0.5	1	0.020702	0.050447	0.029745	39239.2953	1
0.52	0.5	2	0.021471	0.051789	0.030318	9805.9962	1
0.52	0.51	1	0.020794	0.050447	0.029654	39207.8998	1
0.53	0.5	3	0.022765	0.054033	0.031268	4355.3852	1
0.53	0.51	2	0.021612	0.051792	0.03018	9794.2229	1
0.53	0.52	1	0.020973	0.050448	0.029475	39145.1087	1
0.54	0.5	4	0.0246	0.057187	0.032587	2447.6712	1
0.54	0.51	3	0.022959	0.05404	0.031081	4348.4084	1
0.54	0.52	2	0.02184	0.051795	0.029955	9774.6007	1
0.54	0.53	1	0.021235	0.050449	0.029214	39050.9222	1
0.55	0.5	5	0.027002	0.061264	0.034261	1564.6721	1
0.55	0.51	4	0.024855	0.057202	0.032346	2442.7657	1
0.55	0.52	3	0.023241	0.054049	0.030808	4337.9432	1
0.55	0.53	2	0.022148	0.0518	0.029652	9747.1296	1
0.55	0.54	1	0.021569	0.050451	0.028881	38925.3401	1
0.56	0.5	6	0.030001	0.066278	0.036277	1085.0181	1
0.56	0.51	5	0.027327	0.061291	0.033964	1560.9046	1
0.56	0.52	4	0.025199	0.057222	0.032022	2435.8979	1
0.56	0.53	3	0.023603	0.054062	0.03046	4323.9897	1
0.56	0.54	2	0.022525	0.051807	0.029281	9711.8097	1
0.56	0.55	1	0.021964	0.050452	0.028488	38768.3625	1
0.57	0.5	7	0.033634	0.07225	0.038616	795.8017	1
0.57	0.51	6	0.030406	0.066325	0.035918	1081.9657	1
0.57	0.52	5	0.027743	0.061328	0.033585	1555.8813	1
0.57	0.53	4	0.025622	0.057248	0.031626	2427.0679	1

1200 additional rows in `Table03e_asymptotic_vs_exact_n20.csv`.

S3. Benchmark confirmation (Supplementary Table 8)

The study confirmation rule is that the lower Clopper–Pearson 95% limit for p is at least 0.95. After 20/20 positives that limit remains below 0.95. Table 8 gives the exact probability of meeting the rule when the true detection probability is 0.90, 0.95, or 0.99.

n	p_true	P_CP_lower_ge_0.95	CP_lower_after_n_of_n
20	0.9	0.0	0.8316
20	0.95	0.0	0.8316
20	0.99	0.0	0.8316
53	0.9	0.0	0.9328
53	0.95	0.0	0.9328
53	0.99	0.0	0.9328
60	0.9	0.0	0.9404
60	0.95	0.0	0.9404
60	0.99	0.0	0.9404

S4. Qualitative pass rules (Supplementary Tables 9b and 9c)

Table 9b: smallest n such that P(X k | p) 0.95 for k {19, 18, 17}. Table 9c: pairs in which both arms can pass 19/20 while exact Fisher power remains low.

k	p_pct	target_P_pass	min_n	pass_rule_at_min_n	P_pass_at_min_n
19	90	0.95	24	19/24	0.972342
18	90	0.95	23	18/23	0.977392
17	90	0.95	22	17/22	0.981784
19	95	0.95	22	19/22	0.977818
18	95	0.95	21	18/21	0.981119
17	95	0.95	20	17/20	0.984098
19	99	0.95	20	19/20	0.983141
18	99	0.95	19	18/19	0.984726
17	99	0.95	18	17/18	0.986244

pair	n_per_arm	pass_rule	P_pass_arm_a	P_pass_arm_b	P_both_arms_pass	exact_fisher_power	miss_beta_fisher
90/80	20	19/20	0.391747	0.069175	0.027099	0.072484	0.927516
95/85	20	19/20	0.73584	0.175558	0.129182	0.069645	0.930355
90/70	20	19/20	0.391747	0.007637	0.002992	0.241679	0.758321
95/90	20	19/20	0.73584	0.391747	0.288263	0.016712	0.983288

At p = 0.95, P(X 19 | n = 20) = 0.7358.

S5. Protocol audit ledger (Supplementary Table 7b)

Convenience sample, not a prevalence survey. Mapped claims in the ledger still use internal labels E1/E2/E3 (benchmark confirmation / two-arm / qualitative pass). Main text uses the descriptive names.

Compact summary

no	audit_id	evidence_tier	mapped_claim	archived_n	guidance_n_star	n_ratio_vs_oc20	oc_metric	oc_value_at_archived_
1	analyseit_ep17_estradiol	A	qualitative pass	20	21	1.05	E3_pass_probability	0.7358
2	analyseit_ep17_probit	A	two-arm	20	116	5.8	Fisher_power	0.0725
3	zenodo_mus_multiplex_qpcr	A	two-arm	20	199	9.95	Fisher_power	0.0725
4	zenodo_mus_wt_qpcr	A	two-arm	20	116	5.8	Fisher_power	0.0725
5	mendeley_carobene_lombardy_cpa	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0
6	bioc_msstats_lobd_cptac	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0
7	zenodo_radioimmuno_precision	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0
8	zenodo_radioimmuno_linearity	A	benchmark confirmation	20			E1_CP_confirm_probability	0.0
9	zenodo_rma_qualification	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0
10	plos_bzdok2024_trec_end	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0
11	plos_bzdok2024_trec_excl_1to10	A-resp	benchmark confirmation	14	53	3.79	E1_CP_confirm_probability	0.0
12	plos_bzdok2024_trec_excl_1to20	A-resp	benchmark confirmation	12	53	4.42	E1_CP_confirm_probability	0.0
13	plos_bzdok2024_trec_[illegible]_1to100	A-resp	benchmark confirmation	9	53	5.89	E1_CP_confirm_probability	0.0
14	jcm_hansen2020_abbott	A-pub	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0
15	plos_lim2021_mcatarrhalis	A-pub	benchmark confirmation	50	53	1.06	E1_CP_confirm_probability	0.0
16	scirep_jiang2025_mp	A-pub	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0

Verbatim quotes (truncated display; full CSV in the table folder)

audit_id	evidence_tier	mapped_claim	archived_guidance_n	ratio_vs_20	stat_metric	calc_value	target_archived_n	claimed_source	protocol_quote
analyseit_A	ep17_estradiol	qualitative pass	20	21	1.05	E3_pass_probability	0.7358	p=0.95; k/n=19/20	CLSI EP17-A2 (2012) Sec 7.3; mapped to Analyse-it Appendix A estradiol workbook context	CLSI EP17-A2 Section 7.3: a minimum of 20 replicates at the LoD concentration; for a 95% detection claim, at least 19 of 20 replicates should be detected.

audit_id	evidence_tier	mapped_claim	achieved_n	guidance_n	n_ratio_vs_20	stat_metric	oc_value	target_archetype	quote_source	protocol_quote
analyseit_ep17_probit	A	two-arm	20	116	5.8	Fisher_power	0.0725	p_A=0.90, p_B=0.80 (reference lot contrast near LoD)	Analyse-it EP17-A2-Example-4.xlsx, Data sheet header	CLSI EP17-A2 - Appendix C: concentration × replicate detect outcomes for Lot 1, Lot 2, and Lot 3 (hit-rate probit example workbook).

audit_id	evidence_tier	mapped_claim	archived_n	guidance_n	n_ratio_vs_20	stat_metric	oc_value_at_archived_n	target_claim	quote_source	protocol_quote
zenodo_mus_multiplex_qpcr	A	two-arm	20	199	9.95	Fisher_power	0.0725	p_A=0.99; p_B=0.80 (manuscript reference pair at folklore n)	Piaggio A. Zenodo 12609243; multiplex LOD-LOQ CSV	Dataset: Mus WT and tCRISPR multiplex LOD-LOQ (Cq vs standard quantity); steep hit-rate curve near LoD used for lot/target contrast planning.

audit_id	evidence_tier	mapped_claim	derived_n	guidance_n	n_ratio_vs_20	stat_metric	oc_value	target_archetype	claimed_n_source	protocol_quote
zenodo_mus_wt_qpcr	A	upper-arm	20	116	5.8	Fisher_power	0.0725	1225-cell E2 median n* (Table 7); steep hit-rate curve near LoD (Fig. 2A)	Piaggio A. Zenodo 12609243; WT LOD-LOQ CSV	Dataset: Mus WT assay LOD-LOQ (Cq vs copy number); hit-rate series by standard quantity near LoD.

audit_id	evidence_tier	mapped_claim	archived_n	guidance_n_star	n_ratio_vs_20	oc_metric	oc_value	target_achieved	quote_source	protocol_quote
mendeley_carobene_commutability_eqa	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0	p0=0.95 benchmark confirmation	Carobene A. et al. Clin Chim Acta 2026; Mendeley 10.17632/r4szpw8sn5	Ten laboratories measured ALT, AST, ALP, CK, LDH, creatinine, and calcium in EQA materials with raw replicate-level results (commutability study).

audit_id	evidence_tier	mapped_archetype	claimed_n	guidance_n	n_ratio_vs_20	star_metric	oc_value	target_archetype	claimed_n_source	protocol_quote
bioc_msstats_lobd_cptac	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0	p0=0.95 benchmark confirmation	Galitzine et al. MCP 2018; Bioconductor MSstatsLOBD::raw_data	CPTAC peptide spike-in calibration: blank (Concentration=0) and concentration grid for MSstatsLOBD limit-of-blank estimation.
zenodo_radioimmuno_precision	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0	p0=0.95 benchmark confirmation	Malkamäki et al. Anal Chem 2025; Zenodo 12664367 Precision.txt	Precision.txt: per-sample replicate mean/SD/CV across radioimmuno-dot-blot pools (LoQ-style precision profile).

audit_id	evidence_tier	mapped_claim	achieved_n	guidance_n	ratio_vs_20	star_metric	oc_value	target_archetype	archived_source	protocol_quote
zenodo_radioimmuno_linearity	A	benchmark confirmation	20			E1_CP_confirm_probability	0.0	p0=0.95 (continuous proxy; no hit-rate plan in file)	Malkamäki et al.; Zenodo 12664367 Linearity_1.txt	Linearity_1.txt: concentration (ng) × DPM replicate measurements across the linear range.
zenodo_rmn_qualification	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0	p0=0.95 benchmark confirmation	Sawant et al. Zenodo 10.5281/zenodo.17108929	RMN assay qualification dataset: replicate ID50 by dilution for microneutralization LoD-adjacent series.

audit_id	evidence_tier	mapped_claim	claimed_n	derived_guidance_nn	ratio_vs_20	oc_metric	oc_value	target_claimed	archived_source	protocol_quote
plos_bzdok2024_trec_lod	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0	p0=0.95 benchmark confirmation	Bzdok J et al. PLOS ONE 2024; Methods §TREC limit of detection (doi:10.1371/journal.pone.0306	To determine the limit of detection, we analysed a serial dilution of plasmid DNA containing the TREC locus; we were able to reliably detect 5 TREC copies per PCR reaction.

audit_id	evidence_tier	mapped_claim	derived_guidance_n	n	n_ratio_vs_20	oc_metric	oc_value	target_archived	claimed_source	protocol_quote
plos_bzdok2024_trec_cxcl14_1to100	A-hosp	benchmark confirmation		53	3.79	E1_CP_0.0	confirm_probability	p0=0.95; empirical 8/14 (57%) from replicate CSV	Bzdok J et al. PLOS ONE 2024; S3 Dataset (10.1371/journal.pone.0306329.s003); CC BY 4.0	Limit of detection raw data for TREC compared for CXCE-buffer extractions (S3 Dataset sheet 3; per-replicate Cp).
plos_bzdok2024_trec_cxcl12_1to1000	A-hosp	benchmark confirmation		53	4.42	E1_CP_0.0	confirm_probability	p0=0.95; empirical 1/12 (8%) from replicate CSV	Bzdok J et al. PLOS ONE 2024; S3 Dataset (10.1371/journal.pone.0306329.s003); CC BY 4.0	Sub-LoD dilution panel at 1:1000 with undetermined Cp recorded as non-detect (S3 Dataset sheet 3).

audit_id	evidence_tier	mapped_claim	archived_n	guidance_n	ratio_vs_20	metric	oc_value	target_archived	quoted_source	protocol_quote
plos_bzdok2024_trec_magnapure_lod100	A-hosp	benchmark_confirmation	9	53	5.89	E1_CP_confirm_probability	0.0	p0=0.95; empirical 8/9 (89%) from replicate CSV	Bzdok J et al. PLOS ONE 2024; S3 Dataset (10.1371/journal.pone.0306329.s003); CC BY 4.0	Limit of detection raw data for TREC compared for MagNA Pure 96 extractions (S3 Dataset sheet 3; per-replicate Cp).

audit_id	evidence_tier	mapped_claim	derived_n	guidance_n_star	ratio_vs_20	metric	oc_value	target_archived	source	protocol_quote
jcm_hansen2020_abbott	A-pub	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0	p0=0.95; published 16/20 (80%)	Hansen G et al. J Clin Microbiol 2020; CDC/clinical laboratory comparison	Direct comparison of SARS-CoV-2 analytical limits of detection across seven molecular assays (Table 1: detected/tested at stated concentration).

audit_id	evidence_tier	mapped_archetype	claimed_n	derived_guidance_nn	ratio_vs_20	oc_metric	oc_value	target_achieved	claimed_n_source	protocol_quote
plos_lim2021_pub	A-pmc_clinical	benchmark confirmation	50	53	1.06	E1_CP_confirm_probability	0.0	p0=0.95; published 11/50 (22%)	Lim HJ et al. PLOS ONE 2021; Soonchunhyang University hospital-affiliated clinical validation	Analytical sensitivity and LOD were estimated with 50 replicates of positive control bacteria at six different concentrations (Table 3).

audit_id	evidence_tier	mapped_archived_guidance	claimed_n	n	ratio_vs_20	stat_metric	oc_value	target_archived	source	protocol_quote
scirep_jiang2025	A	benchmark confirmation	20	53	2.65	E1_CP_confirm_probability	0.0	p0=0.95; published 19/20 (95%)	Jiang Y et al. Sci Rep 2025; Shaanxi Provincial Hospital clinical validation	Each dilution was assessed using the established assay in 20 replicates. The limit of detection (LoD) was determined by probit analysis, defined as the concentration detectable with 95% probability (Table 1).

S6. Worked example

Archived EP17 Section 7.3 language (Analyse-it estradiol workbook annotation): a minimum of 20 replicates at the LoD concentration, and for a 95% detection claim at least 19 of 20 should be detected.

Locked claim	n at conventional 20	n* under Table 1 targets
Qualitative pass, k held at 19, p = 0.95	P(pass) = 0.736	21
Benchmark confirmation, p = 0.95	CP lower limit after 20/20 < 0.95; P(confirm) = 0	53
Two-arm 90%/80%	exact Fisher power 0.07	199

The audit used the qualitative-pass mapping for that sentence.

S7. Reproducibility

Python 3.10; lod_guide v0.5.0. Tables in this file were regenerated by `scripts/build_online_supplement.py`. Main-text figures remain those in `manuscript/figures/`.